

```
[varnikakomali@VARNIKAs-MacBook-Air ~ % sw_vers
ProductName:      macOS
ProductVersion:   26.5.1
BuildVersion:     25F80
varnikakomali@VARNIKAs-MacBook-Air ~ %
```

```
[varnikakomali@VARNIKAs-MacBook-Air ~ % gcc --version
Apple clang version 21.0.0 (clang-2100.1.1.101)
Target: arm64-apple-darwin25.5.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin
varnikakomali@VARNIKAs-MacBook-Air ~ %
```

2.

A.

```
Last login: Sat Aug  1 13:48:48 on ttys000
[varnikakomali@VARNIKAs-MacBook-Air ~ % pwd
/Users/varnikakomali
```

B.

```
[varnikakomali@VARNIKAs-MacBook-Air ~ % ls
Applications      Music
Desktop           OSLab
Documents         package-lock.json
Downloads         Pictures
EnergyProbabilityProject  Public
installed_apps.txt      PyCharmMiscProject
Library           PycharmProjects
Movies
varnikakomali@VARNIKAs-MacBook-Air ~ %
```

C.

```
[varnikakomali@VARNIKAs-MacBook-Air ~ % cd Desktop
varnikakomali@VARNIKAs-MacBook-Air Desktop %
```

D.

```
[varnikakomali@VARNIKAs-MacBook-Air Desktop % mkdir os_lab  
varnikakomali@VARNIKAs-MacBook-Air Desktop %
```

E.

```
[varnikakomali@VARNIKAs-MacBook-Air Desktop % cd os_lab && mkdir temp_dir && rmdir temp_dir  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```



F.

```
[varnikakomali@VARNIKAs-MacBook-Air os_lab % cp sample.txt copy_sample.txt  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```



G.

```
[varnikakomali@VARNIKAs-MacBook-Air os_lab % mv copy_sample.txt moved_sample.txt  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```

H.

```
[varnikakomali@VARNIKAs-MacBook-Air os_lab % rm moved_sample.txt  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```

I.

```
[varnikakomali@VARNIKAs-MacBook-Air os_lab % echo "Testing cat command" > sample.txt && cat sample.txt  
Testing cat command  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```

J.

```
[varnikakomali@VARNIKAs-MacBook-Air os_lab % echo Hello World  
Hello World  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```

K.

```
[varnikakomali@VARNIKAs-MacBook-Air os_lab % find . -name sample.txt  
./sample.txt  
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```

L.

```
887 cd Desktop\n
888 mkdir OS_Practice\n
889 mkdir temp_folder\nrmkdir temp_folder\n
890 clear
891 pwd
892 4ls
893 ls
894 mkdir test_folder && cd test_folder
895 pwd
896 ls
897 cd Desktop
898 mkdir os_lab
899 cd os_lab && mkdir temp_dir && rmdir temp_dir
900 cp sample.txt copy_sample.txt
901 touch sample.txt
902 cp sample.txt copy_sample.txt
903 mv copy_sample.txt moved_sample.txt
904 rm moved_sample.txt
905 cat sample.txt
906 echo "Testing cat command" > sample.txt && cat sample.txt
907 echo "Hello from Linux Shell\n\necho 'Hello from Linux Shell'
908 echo Hello World
909 find . -name sample.txt
varnikakomali@VARNIKAs-MacBook-Air os_lab %
```

```
varnikakomali@VARNIKAs-MacBook-Air os_lab % echo $SHELL
/bin/zsh
```

3.

4.

```
Last login: Tue Aug 11 22:09:28 on ttys000
varnikakomali@VARNIKAs-MacBook-Air ~ % >....

# 2. Create the Makefile
cat << 'EOF' > Makefile
CC = gcc
CFLAGS = -Wall
TARGET = app

all: $(TARGET)

$(TARGET): main.c
    $(CC) $(CFLAGS) main.c -o $(TARGET)

run: all
    ./$(TARGET)

clean:
    rm -f $(TARGET)
EOF

# 3. Test Makefile targets: clean, compile (make), and run (make run)
make clean
make
make run
zsh: command not found: #
zsh: command not found: #
zsh: no matches found: (make),
rm -f app
gcc -Wall main.c -o app
./app
=====
Shell Architecture & Build Automation
Flow: User -> Shell -> Kernel -> Hardware
=====
varnikakomali@VARNIKAs-MacBook-Air ~ %
```


5.

```
gcc -Wall fork_demo.c -o fork_demo
./fork_demo
=== Starting Process Creation Demo ===
Original Parent PID: 2187
[PARENT] Hello from Parent Process!
[PARENT] Parent PID: 2187 | Created Child PID: 2188
[CHILD] Hello from Child Process!
[CHILD] Child PID: 2188 | Parent PID (PPID): 2187
```

6.

```
gcc -Wall exec_demo.c -o exec_demo
./exec_demo
[CHILD] About to execute 'ls -l' using execvp()...
total 448
drwxr-xr-x  6 varnikakomali staff   192 11 Aug 22:09 2520090010_OSSP
-rw-r--r--  1 varnikakomali staff    62  7 Aug 13:54 addition.c
-rwxr-xr-x  1 varnikakomali staff 33424 11 Aug 22:13 app
drwx-----@ 5 varnikakomali staff   160  5 Mar 13:46 Applications
drwx-----@ 9 varnikakomali staff   288 10 Aug 22:15 Desktop
drwx-----@ 10 varnikakomali staff   320  6 Aug 11:17 Documents
drwx-----@ 16 varnikakomali staff   512 11 Aug 18:22 Downloads
drwxr-xr-x@  4 varnikakomali staff   128 27 Jul 08:33 EnergyProbabilityProject
-rwxr-xr-x  1 varnikakomali staff 33720 11 Aug 22:15 exec_demo
-rw-r--r--  1 varnikakomali staff   701 11 Aug 22:15 exec_demo.c
-rwxr-xr-x  1 varnikakomali staff 33608 11 Aug 22:15 fork_demo
-rw-r--r--  1 varnikakomali staff   717 11 Aug 22:15 fork_demo.c
-rw-r--r--  1 varnikakomali staff 92837 15 Dec  2025 installed_apps.txt
drwx-----@ 91 varnikakomali staff  2912 23 Jul 14:47 Library
-rw-r--r--  1 varnikakomali staff   286 11 Aug 22:13 main.c
-rw-r--r--  1 varnikakomali staff   157 11 Aug 22:13 Makefile
drwx-----  6 varnikakomali staff   192  2 Oct  2025 Movies
drwx-----+ 6 varnikakomali staff   192  2 Oct  2025 Music
drwxr-xr-x  5 varnikakomali staff   160  3 Aug 20:23 OS_LAB
drwxr-xr-x  6 varnikakomali staff   192 31 Jul 22:38 OSLab
-rw-r--r--  1 varnikakomali staff    92  6 Jun 11:38 package-lock.json
drwx-----+ 5 varnikakomali staff   160 27 Sep  2025 Pictures
drwxr-xr-x  6 varnikakomali staff   192  3 Aug 12:23 practical
drwxr-xr-x  2 varnikakomali staff    64  3 Aug 11:10 Practocal
drwxr-xr-x+  4 varnikakomali staff   128 27 Sep  2025 Public
drwxr-xr-x@ 10 varnikakomali staff   320 25 Jul 12:35 PyCharmMiscProject
drwxr-xr-x@  3 varnikakomali staff    96 13 Apr 09:04 PycharmProjects
[PARENT] Child finished execution. Returned to Parent.
varnikakomali@VARNIKAS-MacBook-Air ~ %
```

7.

```
gcc -Wall wait_demo.c -o wait_demo
./wait_demo
=== Parent-Child Synchronization Demo ===
[PARENT] Parent (PID: 2213) waiting for Child (PID: 2214)...
[CHILD] Child (PID: 2214) is performing work...
[CHILD] Child work complete. Exiting with status 42.
[PARENT] Child PID 2214 terminated normally with exit status: 42
[PARENT] Synchronization complete. Parent exiting.
varnikakomali@VARNIKAS-MacBook-Air ~ %
```

8.

```
# 2. Compile the program
gcc -Wall pstree_demo.c -o pstree_demo

# 3. Run the program in the background and immediately inspect via ps
./pstree_demo &
sleep 1
ps -o pid,ppid,command | grep pstree_demo | grep -v grep
wait
zsh: command not found: #
zsh: command not found: #
zsh: command not found: #
[1] 2227
[ROOT] Main Parent PID: 2227
Process Alive -> PID: 2227 | PPID: 2141
Process Alive -> PID: 2230 | PPID: 2227
Process Alive -> PID: 2229 | PPID: 2227
Process Alive -> PID: 2231 | PPID: 2229
  2227  2141 ./pstree_demo
  2229  2227 ./pstree_demo
  2230  2227 ./pstree_demo
  2231  2229 ./pstree_demo
[1] + done      ./pstree_demo
varnikakomali@VARNIKAS-MacBook-Air ~ %
```

9.

```
# 2. Compile and execute
gcc -Wall syscall_trace.c -o syscall_trace
./syscall_trace
zsh: command not found: #
zsh: command not found: #
=====
  System Call Tracing & Execution Demo
=====

[SYSCALL] open("trace_output.txt") -> Returned File Descriptor: 3
[SYSCALL] write(fd=3, bytes=58) -> Success
[SYSCALL] close(fd=3) -> Status: 0

[SYSCALL] open("trace_output.txt", O_RDONLY) -> fd: 3
[SYSCALL] read(fd=3, bytes=58) -> Content:
--> System Call Tracing: open, read, write, close successful!
[SYSCALL] close(fd=3) -> Closed file cleanly.
varnikakomali@VARNIKAS-MacBook-Air ~ %
```

10.

```
# 2. Compile the Mini Shell
gcc -Wall mini_shell.c -o mini_shell

# 3. Run the Mini Shell
[./mini_shell
zsh: command not found: #
zsh: command not found: #
zsh: command not found: #
=====
      Welcome to Mini Shell!
Type commands (e.g. ls, pwd, date, whoami)
Type 'exit' to quit.
=====
mini-shell [PID:2256]> ls
[CHILD] Executing 'ls' | Child PID: 2258 | Parent PID: 2256
2520090010_OSSP          installed_apps.txt          practical
addition.c              Library                    Practocal
app                     main.c                    pstree_demo
Applications            Makefile                 pstree_demo.c
Desktop                 mini_shell               Public
Documents               mini_shell.c             PyCharmMiscProject
Downloads               Movies                   PycharmProjects
EnergyProbabilityProject Music                     syscall_trace
exec_demo               OS_LAB                  syscall_trace.c
exec_demo.c             OSLab                   trace_output.txt
fork_demo               package-lock.json        wait_demo
fork_demo.c             Pictures                 wait_demo.c
[PARENT] Child PID 2258 finished execution.

mini-shell [PID:2256]> pwd
[CHILD] Executing 'pwd' | Child PID: 2259 | Parent PID: 2256
/Users/varnikakomali
[PARENT] Child PID 2259 finished execution.

mini-shell [PID:2256]> date
[CHILD] Executing 'date' | Child PID: 2263 | Parent PID: 2256
Tue Aug 11 22:22:39 IST 2026
[PARENT] Child PID 2263 finished execution.

mini-shell [PID:2256]>
mini-shell [PID:2256]> whoami
[CHILD] Executing 'whoami' | Child PID: 2264 | Parent PID: 2256
varnikakomali
[PARENT] Child PID 2264 finished execution.

mini-shell [PID:2256]> exit
Exiting Mini Shell. Goodbye!
varnikakomali@VARNIKAs-MacBook-Air ~ %
```